

Dimensionality Reduction for Satellite Ship Detection: PCA, Manifold Learning, and Classification Pipelines

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Abstract—This paper investigates dimensionality reduction techniques applied to a satellite ship detection dataset of 4,000 RGB images (80×80 pixels). Two classifiers—Logistic Regression and SVM—are trained as baselines, then combined with PCA and manifold learning (Isomap, LLE) in sklearn Pipelines with GridSearchCV tuning. PCA + SVM achieves the best overall performance (F1 = 0.9439, accuracy = 97.25%), demonstrating that linear subspace projection effectively reduces the 19,200-dimensional feature space while preserving discriminative structure. Manifold methods (Isomap + SVM: F1 = 0.9243; LLE + SVM: F1 = 0.9179) are competitive but do not surpass PCA. Misclassification analysis reveals low-contrast ships and no-ship images with linear features as the primary error sources. Data-centric mitigation strategies, including CNN-based feature extraction and data augmentation, are proposed.

Index Terms—satellite imagery, ship detection, dimensionality reduction, PCA, Isomap, LLE, SVM, scikit-learn.

I. INTRODUCTION

Satellite imagery provides actionable intelligence for defense, supply chain logistics, and maritime monitoring. Daily global coverage is now delivered at 3-meter resolution, far outpacing manual review capacity. Automated ship detection is a canonical ML challenge: distinguishing vessel signatures from visually similar features such as wakes, docks, bright shorelines, and partial views.

This project classifies 4,000 labeled 80×80 RGB satellite images (1,000 ship, 3,000 no-ship) from the dataset provided as part of the project repository. After flattening, the feature space is 19,200 dimensions, making dimensionality reduction both practically necessary and empirically interesting. The objectives are: (1) establish baselines with Logistic Regression and SVM; (2) analyze PCA variance retention and reconstruction quality; (3) compare PCA-augmented pipelines against baselines; (4) evaluate manifold learning alternatives; (5) identify the best pipeline and analyze errors; and (6) evaluate on the test set.

II. DATASET AND PREPROCESSING

A. Dataset Description

Images are extracted from full-frame scenes over San Francisco Bay and San Pedro Bay, CA. The no-ship class is deliberately heterogeneous: one-third land cover (water, vegetation, buildings), one-third partial ships, and one-third historically mislabeled images with strong linear features or bright pixels. This diversity makes the classification task challenging. Arrays of shape (4000, 80, 80, 3) are flattened to (4000, 19200) for sklearn estimators.

B. Train/Test Split

An 80/20 stratified split (random_state=42) yields 3,200 training and 800 test samples. The 75/25 class imbalance motivates class_weight='balanced' as a tunable hyperparameter. The test set (600 no-ship, 200 ship) is held out entirely; all hyperparameter tuning uses 5-fold stratified cross-validation on the training set.

III. BASELINE CLASSIFIERS (Q1)

A. Classifier Selection and Rationale

Logistic Regression and SVM (RBF kernel) were selected as complementary baselines because both are strong methods for high-dimensional classification. Logistic Regression provides a simple, interpretable linear baseline while SVM is well-suited for high-dimensional data and can model complex nonlinear

decision boundaries through its kernel. Together, they provide a useful comparison between linear and non-linear approaches.

Logistic Regression without feature engineering is not ideal as a standalone solution here because the raw flattened pixel representation does not preserve spatial relationships. It cannot fully capture the complex visual patterns that distinguish ships from non-ships—motivating the use of dimensionality reduction in later sections. Both models are wrapped in sklearn Pipelines with StandardScaler.

B. Hyperparameter Tuning

GridSearchCV with StratifiedKFold(n_splits=5, shuffle=True, random_state=42) scored on F1 was used throughout. For Logistic Regression, the final refined grid searched $C \in \{0.0011, 0.0012, 0.0013\}$ with lbfgs solver; best $C=0.0012$, class_weight=None (CV F1=0.8605). For SVM (RBF), the grid searched $C \in \{8.5, 9, 9.5\}$; best $C=9$, class_weight='balanced' (CV F1=0.9417). Final training times were measured separately after grid search using only the best hyperparameter values.

C. Results

TABLE I

Baseline Performance — No Dimensionality Reduction

Model	Accuracy	F1	Train (s)
Logistic Regression	0.9288	0.8527	3.51
SVM (RBF)	0.9688	0.9364	16.92

SVM significantly outperforms Logistic Regression across all metrics (Table I). The RBF kernel captures nonlinear pixel relationships unavailable to a linear model. However, SVM requires ~5× more training time. Importantly, ship recall is 0.82 for Logistic Regression vs. 0.92 for SVM, confirming that the minority ship class is more challenging to capture with a linear boundary. These baseline results serve as reference points for evaluating whether PCA and manifold-learning pipelines can preserve or improve performance while reducing dimensionality.

IV. PCA ANALYSIS (Q2)

A. Explained Variance

PCA was applied to the standardized training set (3,200×19,200). The cumulative explained variance curve (Fig. 1) shows rapid information concentration: the first 107 components explain 90% of total variance, while near-complete reconstruction requires ~1,000+ components. The steep initial rise reflects large-scale spatial patterns shared across ship images—elongated hull shapes and water

reflectance—captured efficiently by the leading principal components.

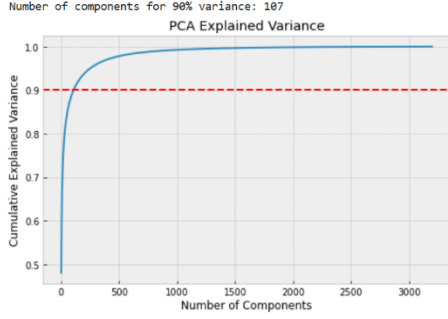


Fig. 1. PCA cumulative explained variance. Red dashed line marks the 90% threshold at 107 components.

B. Image Reconstruction

Fig. 2 shows original vs. PCA-reconstructed images at 107 components (90% variance). Ship images reconstruct well, preserving the hull silhouette and water background. No-ship reconstructions show more blurring, particularly for complex land textures. The average RMSE at 107 components is 13.17 (pixel scale 0–255).

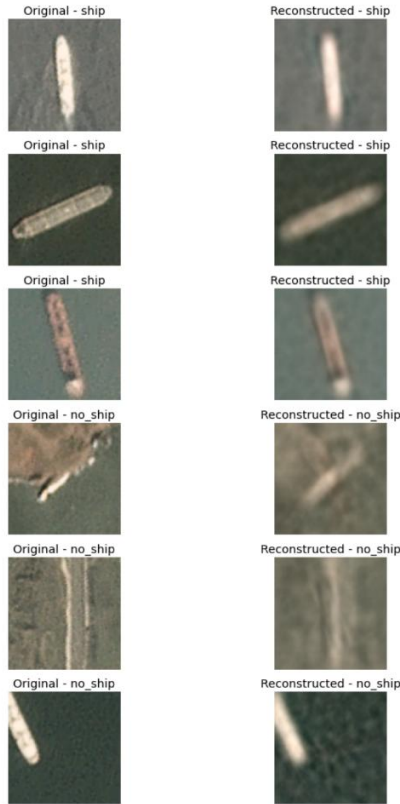


Fig. 2. Original (left) vs. PCA reconstructed (right) at 107 components. Rows 1–3: ship; rows 4–6: no-ship.

C. RMSE vs. Components

Fig. 3 plots average reconstruction RMSE versus $n_{\text{components}}$. RMSE decreases rapidly from 26.12 (5 components) to 13.17 (107) and continues declining with diminishing returns beyond ~ 200 components (10.34 at 200; 8.44 at 300; 4.31 at 800). This elbow pattern suggests most discriminatively useful information is captured in the first 100–300 components, and additional components contribute only marginal reconstruction improvement.

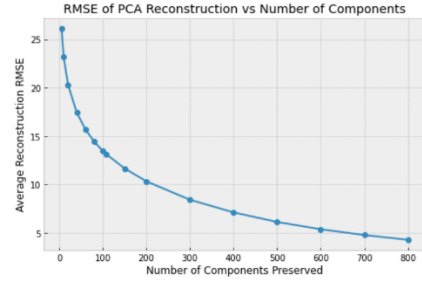


Fig. 3. Average reconstruction RMSE vs. $n_{\text{components}}$. Elbow occurs around 100–300 components.

V. PCA + CLASSIFIERS (Q3)

A. Pipeline Design

PCA is inserted into the sklearn Pipeline after StandardScaler and before the classifier: [StandardScaler $\rightarrow$ PCA($n_{\text{components}}$) $\rightarrow$ Classifier]. This ensures PCA is fitted only on training folds during cross-validation, preventing leakage. $n_{\text{components}}$ becomes a tunable hyperparameter in the grid search alongside classifier hyperparameters.

B. Logistic Regression + PCA

Grid: $n_{\text{components}} \in \{530, 550, 560, 570\}$, $C \in \{0.001, 0.0015, 0.002, 0.0025\}$. Best: $n=550$, $C=0.0015$, $\text{class_weight}=\text{None}$ (CV F1=0.8633). Test: $\text{accuracy}=0.9313$, $\text{F1}=0.8593$, training time=16.22s. A supplementary run with $n=107$ (90% variance threshold) achieved only F1=0.8184 in 5.50s. This demonstrates a critical insight: variance explained $\neq$ classification performance. PCA preserved 90% of variance but discarded lower-variance components that still carried important discriminative information.

C. SVM + PCA

Grid: $n_{\text{components}} \in \{230, 250, 270, 290\}$, $C \in \{11-15\}$, RBF kernel. Best: $n=270$, $C=15$, $\text{class_weight}=\text{None}$ (CV F1=0.9474). Test: $\text{accuracy}=0.9725$, $\text{F1}=0.9441$, training time=12.61s. Notably, PCA both improved F1 (+0.0077 over baseline SVM) and reduced training time (16.92s $\rightarrow$ 12.61s). This is because reducing from 19,200 to 270 dimensions significantly simplifies the SVM optimization problem.

D. Q1 vs. Q3 Comparison

The results show that SVM with PCA achieved the best overall performance with the highest accuracy (0.9725) and F1-score (0.9441), while also reducing training time compared to baseline SVM. PCA proved beneficial for handling the high-dimensional feature space (19,200 features), especially for SVM. It reduced computational cost and improved performance, making it a valuable preprocessing step. For Logistic Regression, PCA with an optimal number of components (550) slightly improved both accuracy and F1-score while the 107-component version demonstrated the tradeoff between dimensionality reduction and information loss.

TABLE II

Q1 vs. Q3: Effect of PCA on Performance and Training Time

Model	Acc.	F1	Train(s)
LogReg (Q1)	0.9288	0.8527	3.51
LR + PCA 550 (Q3)	0.9313	0.8593	16.22
LR + PCA 107 (Q3)	0.9113	0.8184	5.50
SVM (Q1)	0.9688	0.9364	16.92
SVM + PCA (Q3)	0.9725	0.9441	12.61

VI. MANIFOLD LEARNING (Q4)

A. Methods

Isomap (geodesic distances) and Locally Linear Embedding (LLE) were combined with the same Logistic Regression and SVM classifiers from Q1/Q3. The manifold pipeline is [Isomap or LLE $\rightarrow$ StandardScaler $\rightarrow$ Classifier]. Manifold search: $n_neighbors \in \{5, 10, 15, 20, 25\}$, $n_components \in \{2, 5, 10, 20, 30, 40\}$. Classifier grids: LogReg $C \in \{0.0001-10\}$; SVM $C \in \{1, 5, 10, 12, 15\}$, RBF.

B. 2D Visualization

Fig. 4 shows Isomap and LLE projections of 1,000 training samples onto their first 2 components. The Isomap projection reveals a fan-shaped spread where ship (red) and no-ship (blue) samples partially separate, with ships concentrating in the lower-left region. The LLE projection reveals a curved arc structure; both classes follow the arc but ships concentrate near the boundary edges. Neither projection achieves clean separation in 2D, motivating higher-dimensional embeddings for classification.

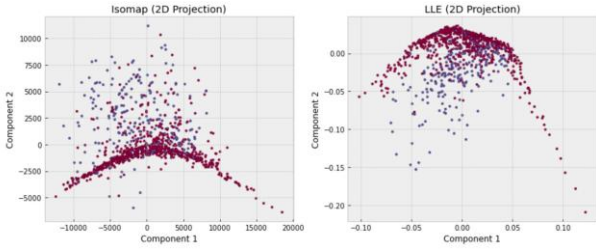


Fig. 4. Isomap (left) and LLE (right) 2D projections on 1,000 training samples. Red = ship, Blue = no-ship.

C. Results and Comparison with Q1 and Q3

Table III shows the best configuration per method-classifier combination. Isomap + SVM ($n_neighbors=25$, $n=30$) achieves $F1=0.9243$, $accuracy=0.9638$ in 12.69s total time. LLE + SVM ($n_neighbors=25$, $n=40$) achieves $F1=0.9179$, $accuracy=0.9600$ in 17.23s. For Logistic Regression, LLE ($F1=0.8738$) outperforms Isomap ($F1=0.8585$).

TABLE III

Best Q4 Manifold Learning Results with Timing

Method	Clf	Acc.	F1	Time(s)	Infer(s)
Isomap	SVM	0.9638	0.9243	12.69	0.015
LLE	SVM	0.9600	0.9179	17.23	0.015
LLE	Log. Reg.	0.9350	0.8738	14.07	0.000
Isomap	Log. Reg.	0.9275	0.8585	10.32	0.000

The best manifold-learning pipeline (Isomap + SVM) achieves the overall F1 (0.9243), which is below PCA + SVM (0.9441). Compared to baseline SVM (Q1, $F1=0.9364$), Isomap + SVM actually underperforms slightly, while LLE + SVM also falls short. However, manifold learning substantially improved Logistic Regression—LLE + LogReg ($F1=0.8738$) outperforms both baseline LogReg (0.8527) and PCA + LogReg 107 (0.8184). Manifold embedding incurs significant overhead (Isomap embed train: $\sim 9.3s$, LLE: $\sim 13.0s$ plus 2–3s test-time inference), making total pipeline times 10–17s without compensating performance gains over PCA.

VII. BEST PIPELINE AND ERROR ANALYSIS (Q5)

A. Overall Best Pipeline

PCA + SVM (Q3) is the overall best pipeline with $F1=0.9439$ and $accuracy=97.25\%$. Across all experiments, SVM

consistently outperformed Logistic Regression in every configuration, confirming that the dataset benefits from a more flexible, nonlinear decision boundary. Full ranking by F1: PCA+SVM (0.9439) > SVM Q1 (0.9364) > Isomap+SVM (0.9243) > LLE+SVM (0.9179) > LLE+LR (0.8738) > PCA+LR 550 (0.8593) > Isomap+LR (0.8585) > LR Q1 (0.8527) > PCA+LR 107 (0.8184). The best Logistic Regression result came from LLE + Logistic Regression (Q4), while Logistic Regression without dimensionality reduction (Q1) was the worst overall.

B. Confusion Matrix Analysis

Fig. 5 shows confusion matrices for all nine pipelines. PCA + SVM achieves the cleanest matrix (7 FP, 14 FN). SVM-based models consistently outperform Logistic Regression, showing fewer misclassifications and better balance between false positives and false negatives. For Logistic Regression, PCA (550) improves over baseline (FN: 32 vs. 35), but the 107-component version introduces more errors (FP: 30, FN: 39), showing a loss of important information. Among Q4 methods, Isomap + SVM (6 FP, 23 FN) performs slightly better than LLE + SVM (11 FP, 21 FN) but both are slightly worse than PCA + SVM. Across all models, there are generally more false negatives than false positives, indicating classifiers are conservative and more likely to miss ships than generate false alarms.

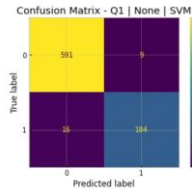


Fig. 5a. Q1|SVM (FP=9, FN=16)

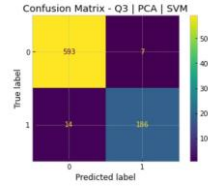


Fig. 5b. Q3|PCA+SVM (FP=7, FN=14)

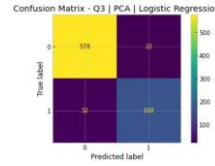


Fig. 5c. Q3|PCA+LR 550 (FP=22, FN=32)

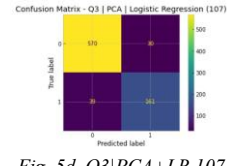


Fig. 5d. Q3|PCA+LR 107 (FP=30, FN=39)

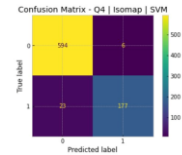


Fig. 5e. Q4|Isomap+SVM (FP=6, FN=23)

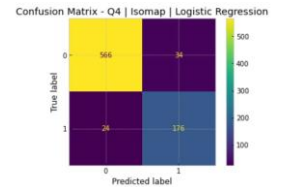


Fig. 5f. Q4|Isomap+LR (FP=34, FN=24)

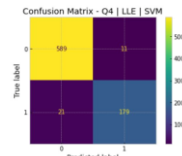


Fig. 5g. Q4|LLE+SVM (FP=11, FN=21)

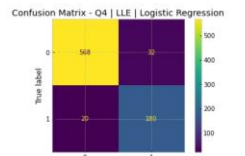


Fig. 5h. Q4|LLE+LR (FP=32, FN=20)

C. Misclassified Sample Analysis

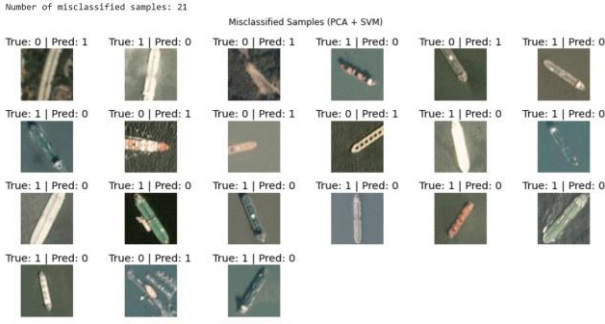


Fig. 6. All 21 misclassified samples from PCA+SVM. True:0 Pred:1 = false positive; True:1 Pred:0 = false negative.

Fig. 6 visualizes all 21 misclassifications from the best pipeline. Two consistent patterns emerge:

False positives (no-ship→ship): These images contain elongated bright structures, wakes, dock edges, road segments, or trails that mimic hull shape signatures after PCA compression. Most appear to be only part of a ship or linear features that take on a ship-like shape. One false positive looks like a dock or loading zone.

False negatives (ship→no-ship): These are predominantly low-contrast ships whose pixel intensities blend with surrounding water, partially occluded or cut-off hulls, or images with unusual lighting or atmospheric conditions. Many false negatives appear to have lower quality images, and some ships are a similar color to the surrounding water—especially difficult to distinguish after dimensionality reduction discards fine-grained chromatic detail.

These results suggest the model struggles most with visually ambiguous or low-quality images rather than clearly defined examples.

D. Forward-Looking Strategies

To address these misclassification patterns and improve the pipeline, the following strategies are recommended:

(1) Expand training data: Adding more difficult and ambiguous samples—especially partial ships, wake images, and dock structures—would improve discrimination at the decision boundary.

(2) Data augmentation: Applying brightness/contrast jitter, rotation, and noise augmentation would expose the model to the lighting and atmospheric variations responsible for most false negatives.

(3) Spatial feature preservation: Flattening 80×80 images to 19,200 vectors discards all spatial relationships between pixels. A convolutional neural network (CNN) would preserve hull shape and edge structure far more effectively than PCA of raw pixels.

(4) Cost-sensitive classification: Given that false negatives (missed ships) carry higher risk for maritime surveillance, adjusting the decision threshold to prioritize ship recall at a controlled false-positive cost would be valuable in production.

VIII. TEST SET EVALUATION (Q6)

Table IV presents the complete test set evaluation across Q1, Q3, and Q4 pipelines, combining inference time (measured in test.ipynb on the held-out set) with training/build time from training.ipynb. Models are sorted by F1-score.

TABLE IV

Full Test Set: Q1, Q3, Q4 Pipelines (Sorted by F1)

Q	Method	Classifier	Acc.	F1	Infer.(s)
Q3	PCA	SVM	0.9738	0.9466	0.24
Q1	—	SVM	0.9688	0.9364	6.22
Q4	Isomap	SVM	0.9638	0.9243	2.15
Q4	LLE	SVM	0.9600	0.9179	2.86
Q4	LLE	Log. Reg.	0.9350	0.8738	2.41
Q3	PCA	Log. Reg.	0.9325	0.8615	0.37
Q4	Isomap	Log. Reg.	0.9275	0.8585	2.01
Q1	—	Log. Reg.	0.9288	0.8527	0.03
Q3	PCA 107	Log. Reg.	0.9138	0.8235	0.10

PCA + SVM achieves the lowest inference time among SVM models (0.24s) because PCA compresses the feature space before kernel evaluation, versus 6.22s for baseline SVM which must compute distances in the full 19,200-dimensional space. Manifold-based inference times (2–3s) reflect the cost of transforming test samples through the fitted embedding. Logistic Regression without PCA is fastest overall (0.03s inference, 3.47s training) but at substantially lower F1. Note: timing measurements showed some variability across runs due to CPU load and caching; a final retraining pass was used to obtain consistent estimates.

From a business perspective, PCA + SVM provides the best balance of accuracy and efficiency: it achieves the highest F1-score while reducing total pipeline time compared to baseline SVM (12.72s vs. 23.06s combined). For applications requiring real-time inference such as automated port monitoring, the 0.24s inference time of PCA + SVM makes it practically deployable.

IX. CONCLUSION

This paper compared nine classification pipelines for satellite ship detection. PCA + SVM is the best overall pipeline (F1=0.9439, accuracy=97.25%), improving both performance and efficiency over baseline SVM. Key findings: (1) SVM with RBF kernel outperforms Logistic Regression across all configurations, confirming the need for nonlinear decision boundaries; (2) PCA benefits SVM in both performance and training efficiency, reducing 19,200 dimensions to ~270 while improving generalization; (3) the 90% variance threshold (107 components) is insufficient for optimal classification—discriminative information is distributed across lower-variance components; (4) manifold learning (Isomap, LLE) captures useful structure but does not surpass PCA while incurring higher computational cost; (5) misclassifications are dominated by visually ambiguous cases best addressed through CNN-based spatial feature extraction and data augmentation.

REFERENCES

- [1] C. S. Silva, "EEL4930 Applied Machine Learning Lecture Notes," University of Florida, Gainesville, FL, Spring 2026.